

Project # 3

Due Date: Sun: 03/09/2019

Points: 100

Make cadence schematic for comparator.

Use 1-stage opamp in the preamplifier with strong-arm latch with $V_{DD} = 1.3V$ supply.

1. Design (20 pts):

Use $C = 1$ pF for storing offset and use output-offset cancellation (preferred). You can design your own pre-amplifier and latch using reference works from IEEE Explore and Class Notes.

Highlight the following in your design choices through analysis (and using block-level simulation to verify):

- Input offset voltage (show through analysis only – dependent on your W and L of your preamp).
- Bandwidth of the preamplifier
- Power of the preamplifier
- Time-constant of the latch
- Leakage current in the latch (leave the supply connected but disable the clock) (in the off-state)

2. Testbench Setup (10 pts)

Generate clocks from V_{DD} to V_{SS} with rise and fall times of 40 ps. Use clock frequency of 1 GHz. Show the testbench schematic clearly with the DC / AC / Tran sources.

3. Observations to report from DC simulation (10 pts)

- Show DC voltage of the preamp in schematic (with proper i/p biases)
- Report DC Gain, 3dB BW and unity-gain bandwidth of the open-loop preamplifier.

4. Observations to report from transient simulation (40 pts) :

- Fig. 1: Show clock waveform of the preamp and latch clearly (5 pts)
- Fig. 2: Show the input, preamp output, latch output with clock waveform clearly (5 pts)
- You will add additional figures to show your performance

In addition to the above figures, please verify and report the following:

- The charge injection (CI) error by measuring V'' 0.1 ns before and 0.1 ns after ϕ_{1a} turns OFF (note ϕ_2 is OFF).
- Reset phase settling time: with $V_{in} = +0.5$ V, measure time for V' and V'' to settle to $< 10mV$ of its final value
- Comparator phase settling time
- With V_{in} set to 50 mV, measure how long it takes for the output to reach $> |1.0 V|$.
- How long does it take to reach $> |1.0 V|$ if the compensation capacitor is not switched (tie gate of compensation FET Q1 to V_{DD}).
- Change clock amplitude to 0.9 V and measure CI error.

5. Observations to report from Monte-Carlo simulation: (20 pts)

- Run monte-carlo simulation with the latch and the preamplifier included. For setting Monte-Carlo, use ADE-XL or ADE-Explorer and use 100 iterations with random mismatch. Your IBM model is Monte-Carlo compatible. Report the variation in the input-offset of the comparator with the latch in operation.